# Chip Canary: Are Korea's Semiconductor Export Figures Already Priced In?

*Release timing, earnings and market reactions, 2016-2026*

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**Abstract.** Korea publishes its chip export figures early, around the 21st and the 1st of each month, while SK Hynix, a chip maker, and Samsung, whose results depend heavily on chips, report their earnings weeks later. If investors already watch these figures, do the two firms' earnings still tell the market anything new? I dated every figure by the day it became public and used first-release values where they exist, rebuilding the first-20-days series myself from 121 Korean customs releases, then compared these exports with the earnings reported later and measured how share prices moved on release days, writing down my predictions before seeing any result. Exports known by the 21st track the earnings published two to five weeks later very closely, and for Samsung the link with revenue appears to have strengthened during the AI boom. Yet share prices move only moderately more than usual on release days, and the 20-day figure does not move markets more than the monthly one. Whether this information is already priced in on earnings day remains open: the data are consistent with the idea, but too few to establish it.

## 1. Introduction

Semiconductors have become the core of South Korea's exports: in the first twenty days of September 2026, they made up nearly half of the total, up from less than a fifth at the end of 2023 (47.8% against 17.6% in the first twenty days of December 2023; Korea Customs Service, 21 September 2026 and 21 December 2023). Samsung and SK Hynix are two of the three firms that dominate high-bandwidth memory, the memory used in artificial intelligence chips (TrendForce, 17 September 2025), a sector now central to the world economy and to international relations alike. Korea's export figures, released around the 21st and the 1st of each month, are therefore closely watched by investors worldwide. This note asks a simple question: if these figures already signal how the two firms are doing, do their quarterly earnings still tell the market anything new? To answer it, the note rebuilds the information calendar, from export release to earnings release, and tests predictions written down before seeing any result.

## 2. Data and the information calendar

*In plain terms: this section explains where the numbers come from and, above all, on which day each one became public, because the whole question is what investors could know, and when.*

Every figure is dated by the day it became public, not by the period it describes. Dating by the period would use information nobody had at the time, a *look-ahead bias* that makes any "leading indicator" look better than it was.

**Sources.** The first-20-days report is "closely watched as an early indicator of global trade" (Bloomberg, republished by Infobae, 21 January 2021). Korea Customs publishes it around the 21st (since June 2016); the trade ministry publishes the full month on the 1st. Earnings come from each firm's first *preliminary results filing* (its first, unaudited announcement of quarterly results): Samsung's early-month estimate, or *guidance*, and SK Hynix's release around the 25th. Prices cover both firms and the KOSPI (the main Korean stock market index).

**First release, not revised data.** Where possible the note uses the *first release*, the figure as first published, before any later correction (*revision*), because it is what the market reacted to. The first-20-days chip export growth was rebuilt from 121 customs releases; the first earnings filings also state the same quarter a year earlier, giving *year-on-year* growth (the change from a year before). The full-month figure is the exception: it comes from the Bank of Korea's revised index.

**The calendar** holds 540 events, each dated by its issuer; releases on a weekend or holiday count for the next *trading day* (day the stock market is open). Monthly dates before 2016 cannot be sourced and are set by rule in a separate extension sample (2003 to 2015). Appendix A1 lists the traps met.

> **Box 1. Why the first release matters: Samsung, third quarter of 2016.** On 7 October 2016, Samsung's first preliminary filing put third-quarter operating profit at about KRW 7.8 trillion. On 12 October 2016, after stopping sales, exchanges and production of the Galaxy Note 7, Samsung issued a revised guidance of about KRW 5.2 trillion (Samsung Newsroom, 12 October 2016; CNBC, 12 October 2016); the detailed results of 27 October 2016 confirmed KRW 5.20 trillion. A revised series records only 5.2, while from 7 to 12 October the market knew only 7.8. A study using revised data would date the bad news wrongly.

**Table 1. Main data sources**

| Series | Source | Coverage used | Version |
|---|---|---|---|
| Chip exports, first 20 days of the month (y/y %), rebuilt from 121 releases | Korea Customs Service | 2016-07 to 2026-09 | First release |
| Chip export value index, full month (USD) | Bank of Korea, ECOS | 2000-01 to 2026-08 | Revised |
| Revenue and operating profit, Samsung and SK Hynix | OpenDART preliminary filings | 2009Q4 / 2010Q1 to 2026Q2 | First release |
| Daily prices: Samsung, SK Hynix, KOSPI | Yahoo Finance | 2000 to 2026-09-30 | Adjusted |

*Full table, release-date sources and robustness series: appendix A0. Main sample of the analysis: 2016 to 2026, sourced dates only.*

## 3. Who knows first? Exports versus earnings

*In plain terms: Korea's chip export figures are public several weeks before Samsung and SK Hynix announce their results, and they move very closely with those results, for both companies.*

**Setting.** The information date t is the first-20-days release for the quarter's last month. At t the market knows about eight ninths of the quarter's exports, combined into one year-on-year growth rate. Earnings are each firm's first release; operating profit, which sometimes changes sign, enters as its year-on-year change divided by revenue a year earlier. Sample: 2016Q3 to 2026Q2, 39 quarters per firm.

**The lead is weeks, not days.** Results follow the first-20-days figure by a median of 35 days for SK Hynix and 16 days for Samsung (Figure 1), just above Samsung's *pre-registered* threshold of 15 days (written down before any result, so the bar could not be moved). From the monthly release on the 1st, Samsung's guidance follows after only 6 days.

**Table 2. Exports known at t and first-release earnings, 2016Q3-2026Q2 (n = 39 per cell)**

| Firm | Earnings measure | r | 95% CI (Fisher) | 95% CI (block bootstrap) | Expected | Outcome |
|---|---|---|---|---|---|---|
| SK Hynix | Revenue growth | 0.95 | 0.90 to 0.97 | 0.82 to 0.98 | very strong | as expected |
| SK Hynix | Operating profit change | 0.94 | 0.89 to 0.97 | 0.79 to 0.98 | medium or strong | stronger than expected |
| Samsung | Revenue growth | 0.90 | 0.81 to 0.95 | 0.65 to 0.94 | medium or strong | stronger than expected |
| Samsung | Operating profit change | 0.92 | 0.86 to 0.96 | 0.82 to 0.96 | weak or medium | stronger than expected |

*Categories: weak < 0.3, medium 0.3 to 0.6, strong 0.6 to 0.8, very strong 0.8 or more, fixed before the results. For the second and third rows the bootstrap interval reaches "strong", so the outcome is conclusive with the Fisher interval only.*

**The link is very strong** (Table 2). The *correlation* (from -1 to 1, how closely two series move together) between exports known at t and the results released two to five weeks later is 0.90 to 0.95, *statistically significant* (very unlikely by chance) even after a *Holm correction*, which makes each test stricter when many are run. The 95% *confidence interval*, the range of values compatible with the data, is given by the standard formula (Fisher) and by a *block bootstrap*, a resampling method that keeps consecutive quarters together.

**No clear gap between firms or measures.** The link was expected to be weaker for Samsung, whose chips are only part of the group, and for operating profit, which also depends on costs. Neither gap is confirmed. A possible reading, not tested here: memory prices drive both sales and margins.

**Structural for SK Hynix; the AI boom may matter for Samsung.** Before 2023, SK Hynix's correlations are unchanged (0.94), while Samsung's revenue correlation falls to 0.72 (95% CI 0.45 to 0.87). This is *descriptive* (shown, not formally tested): the intervals overlap. Results are robust to other checks (appendix A4).

**What this part does not show.** SK Hynix's sales largely are Korea's chip exports, so part of the link is accounting, not forecasting. Whether exports tell the market anything new is the subject of sections 4 and 5.

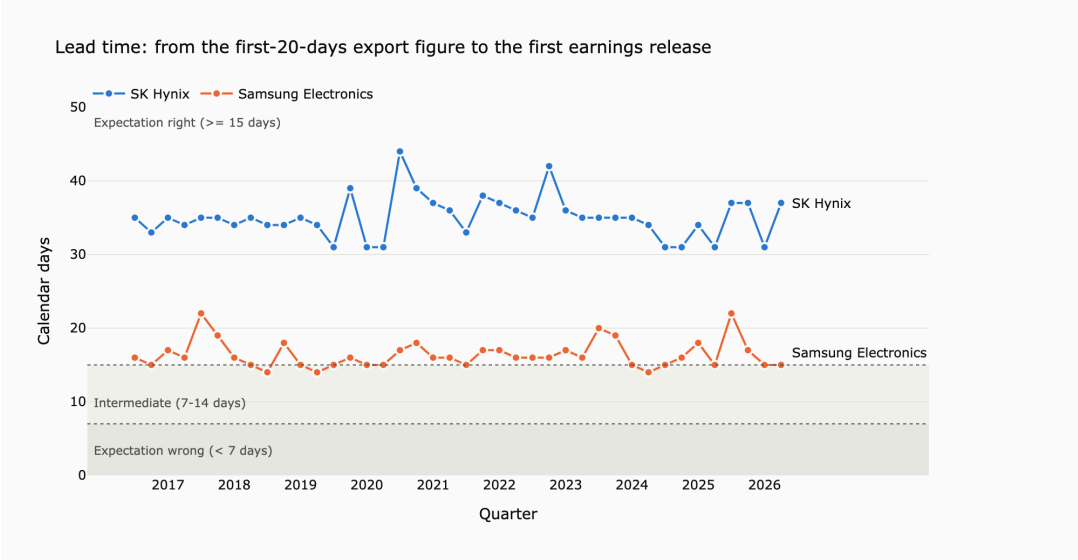



*Figure 1. Lead time from the first-20-days export figure of the quarter's last month to each firm's first earnings release, by quarter, with the pre-registered thresholds.*

## 4. Do markets react on release days?

*In plain terms: on the days these figures are published, share prices move somewhat more than on an ordinary day, but not much more, and the pattern looks like a reaction to general economic news rather than to chips specifically.*

**Method.** An *event study* (MacKinlay, 1997) compares what happens around a known date with what would normally happen. For each release and stock, the normal return is estimated over the preceding 250 trading days; the excess over it is the *abnormal return*, and its sum over the day before, the day of and the day after the release is the *cumulative abnormal return* (CAR). The *reaction ratio* divides the absolute CAR by the average move over three ordinary days of the same year: 1 means an ordinary day, 2 twice as much. Significance comes from a *permutation test* against thousands of random ordinary days, Holm-adjusted over 22 tests. Releases overlapping another type of release are left out.

**Release days are somewhat busier, never much busier** (Figure 2). Moves are 22% (SK Hynix), 35% (Samsung) and 25% (KOSPI) larger than usual after the monthly release, all significant after Holm; 19%, 15% and 3% after the first-20-days release, none significant; 50% and 63% after the two firms' results. No

ratio reaches 2. Without 31 July 2026, the KOSPI's record day, which falls in one monthly window, the monthly estimates barely change but SK Hynix's and the KOSPI's no longer pass Holm.

**The pre-registered contrasts are not found.** The earlier first-20-days figure does not move markets more than the monthly one, and once scaled by its own volatility SK Hynix reacts no more than Samsung or the index. The comparison before and after 2016 cannot be interpreted: rule-based dates, if wrong, measure ordinary days and push the ratio toward 1, the level observed.

**A possible reading, not tested.** Markets may respond to the overall economic signal rather than to chips: the monthly release covers all exports, and the index reacts as much as the chip stocks. Other first-of-month news, such as manufacturing surveys (PMIs), may also play a part.

**Robustness.** Against the US chip index (SOX), SK Hynix's reaction to the first-20-days figure becomes significant (1.30), a fragile result; other checks are in appendix A4.

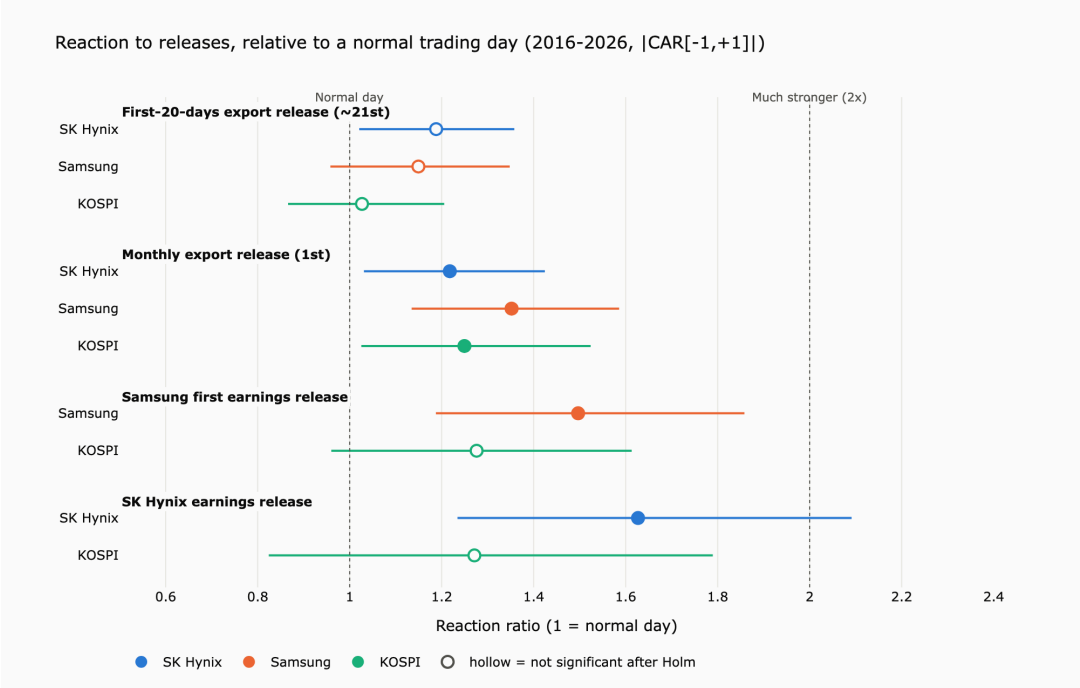



*Figure 2. Reaction ratio by release type and stock (1 = ordinary day), with 95% intervals; hollow markers are not significant after Holm correction.*

## 5. Is the predictable part already priced in?

*In plain terms: if investors had already used the export figures, share prices should not react on results day to the part of the results those figures announced; the data are compatible with this, but they are too few to prove it.*

**The test.** If the part of results that exports announced was *priced in* (already in the share price), the reaction on results day should not depend on it. This predicted part comes from a *regression* (a line fitted between export growth and operating profit) on an *expanding window*, using only quarters already public, to avoid look-ahead. The test asks whether the signed reaction rises with it.

**Compatible with "priced in", but not informative.** Neither slope differs significantly from zero (Figure 3): -1.90 for SK Hynix (95% CI -2.71 to +1.74, 22 quarters), -1.96 for Samsung (-7.09 to +6.18, 31 quarters). But SK Hynix's slope rests on one quarter, 2026Q2 (without it, -0.07), and Samsung's sign depends on 2026 (diagnostics, appendix A4). With such wide intervals, the data fit both a market that had priced the information in and one that still reacted strongly to it.

**What would settle it:** more quarters, release times, and analysts' forecasts, to measure the *surprise* relative to expectations (Andersen, Bollerslev, Diebold and Vega, 2003); such forecasts are commercial and

were not available.

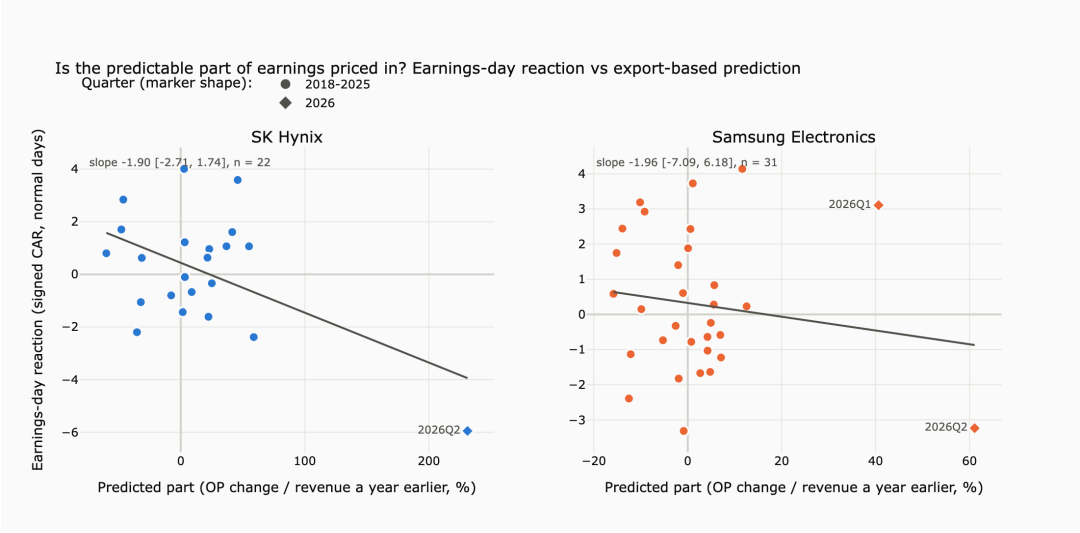



*Figure 3. Earnings-day reaction (signed, in ordinary days) against the part of earnings predicted by exports; 2026 quarters marked.*

## 6. Pre-registration and limitations

*In plain terms: the expected results were written down before any calculation, so that the analysis could not be bent to confirm them; several turned out to be wrong, and this section says which.*

**Why write expectations first.** A study can find something "significant" by trying until it works (*p-hacking*). Here the expectations, thresholds and 32 specifications were written in a dated file before any result, with a log of every run. The expectations themselves were committed to the project's version history on 7 October 2026, before the first result. Some precisions were added later but before the corresponding results (for instance the expected category of each correlation); they are dated in the file, but were committed together with the results, so for them the file's log, not the version history, attests the order. The process was self-administered, not registered with an external body.

**Table 3. Expectations written before the results, and outcomes**

| Expectation | Outcome | Verdict |
|---|---|---|
| Lead of exports over results of 15 days or more | 35 days (SK Hynix), 16 days (Samsung) | Right; just above the threshold for Samsung |
| Very strong link for SK Hynix revenue | r = 0.95 | Right |
| Medium or strong link for SK Hynix operating profit | r = 0.94, very strong | Wrong (stronger) |
| Medium or strong link for Samsung revenue | r = 0.90, very strong (0.72 before 2023) | Wrong (stronger), in range before 2023 |
| Weak or medium link for Samsung operating profit | r = 0.92, very strong | Wrong (stronger) |
| Link stronger for SK Hynix than for Samsung | Difference 0.05 (revenue), 0.01 (profit) | Not confirmed |
| Link stronger for revenue than for operating profit | Difference 0.01 (SK Hynix), -0.03 (Samsung) | Not confirmed |
| First-20-days figure moves prices more than the monthly one | Slightly less, for all three | Wrong |
| Reaction to the monthly figure weaker since 2016 | Stronger, but pre-2016 dates set by rule | Not interpretable |

| Expectation | Outcome | Verdict |
|---|---|---|
| SK Hynix reacts more than Samsung, Samsung more than the index | About equal once scaled by volatility | Wrong |
| Some movement on results day, not much larger than usual | Ratios 1.5 (Samsung), 1.6 (SK Hynix) | Right; for SK Hynix the interval reaches 2 |
| Reaction on results day stronger when the lead is short (exploratory) | Slopes slightly positive, intervals include zero; the lead varies by about 2 days only | Not conclusive |
| Much larger reactions to large surprises (exploratory, monthly release only) | Top 20% surprises vs others: -0.02 (SK Hynix), -0.45 (Samsung), -0.54 (KOSPI) standard deviations | Wrong for Samsung and the index, not conclusive for SK Hynix |
| Information already priced in before results | Slope not different from zero, but driven by 2026 | Open |

*Thresholds: correlation categories weak < 0.3, medium 0.3 to 0.6, strong 0.6 to 0.8, very strong 0.8 or more; a reaction gap counts if it reaches half an ordinary day. All p-values Holm-adjusted within each part, except the two exploratory rows. In the surprise test, the surprise is the full-month growth (revised series) minus the first-20-days growth, and the reaction is counted in the direction of the surprise; no surprise can be measured for the first-20-days release itself, for lack of a published expectation.*

**Limitations.** Revised data for the monthly figure; a short sample dominated by 2023 to 2026; rule-based dates before 2016; and the accounting overlap between SK Hynix's sales and exports. Others are in appendix A7.

## 7. Conclusion

This note documents a contrast: Korea's chip exports anticipate Samsung's and SK Hynix's earnings closely, two to five weeks ahead, yet share prices react only moderately when these figures are released. As chips weigh more heavily in Korean exports, and their link with Samsung's revenue appears to have strengthened, this signal deserves closer attention. Whether it is already priced in remains open; more quarters, the exact time of each release and analysts' forecasts would help settle it. Two extensions seem natural: a comparison with Taiwan, another major chip exporter, and a study of how US–China export controls shape this signal.

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*Analysis in Python (pandas, statsmodels, plotly). Code and drafting assisted by Claude Code (Anthropic); research design, expectations, decisions and final text by the author. All data sources, scripts and the pre-registration log are in the project repository.*

# Appendix

## A0. Data sources (full table)

| Series | Source | Frequency | Coverage used | Version |
|---|---|---|---|---|
| Chip exports, first 20 days of the month (y/y %) | KCS press releases | Monthly, ~21st | 2016-07 to 2026-09 (120 releases) | First release |
| Release dates, full-month exports | MOTIE press releases | Monthly, 1st | 2016-01 to 2026-10 (130); 2003-2015 by rule | Dates only |
| Chip export value index (2020 = 100, USD) | Bank of Korea, ECOS | Monthly | 2000-01 to 2026-08 | Revised (2026 provisional) |
| Revenue and operating profit | OpenDART preliminary filings | Quarterly | Samsung 2009Q4, SK Hynix 2010Q1, to 2026Q2 | First release |
| Revenue and operating profit, final | OpenDART quarterly reports | Quarterly | 2016Q1 to 2026Q2 | Final |
| Exports of integrated circuits (HS 8542) | UN Comtrade | Monthly | 2013-01 to 2025-12 | Revised (cross-check) |
| Prices: Samsung, SK Hynix, KOSPI | Yahoo Finance | Daily | 2000 to 2026-09-30 (SK Hynix from 2003-04-15) | Adjusted |
| KRW/USD, SOX index | Yahoo Finance | Daily | 2003-12 / 2000 to 2026-09 | Robustness only |

*Download dates and series codes are in the project's `data/SOURCES.md`.*

**Checks of the rebuilt first-20-days series.** The semiconductor export growth of the first 20 days was extracted from the attachments (HWP, HWPX and PDF files) of 120 of the 121 customs releases; the June 2016 release gives only integrated circuits and is left missing. The 35 releases published in several formats give the same figure in each, eight values match press reports of the time, and the series follows the full-month ECOS growth closely (correlation 0.977).

**Cross-check of the ECOS index.** Year-on-year growth of the ECOS index tracks UN Comtrade exports of integrated circuits (HS 8542, the customs code for integrated circuits, 2013 to 2025) with a correlation of 0.992, which validates its processing (not the underlying customs data, common to both) and indicates that it is in US dollars, while earnings are in won.

## A1. Building the release calendar: traps found

- **Posting lag (KCS).** The KCS website shows the date a release was posted, which can lag the actual release by up to six days (figures for 1 to 20 May 2026: released on 21 May, posted on 27 May, confirmed by press reports). 104 of the 121 first-20-days releases are dated by the date written at the start of their attachment file names; the other 17 keep the posting date, 16 of which fall on the expected trading day (one unverified case, September 2016).
- **Site migration (MOTIE).** The MOTIE website shows 9 December 2016 for every release before 2016. Monthly dates before 2016 are therefore set by rule (first day of the month); on 2016-2026 the rule falls on the right trading day for all 129 sourced releases with price data.
- **Unit error (Samsung filings).** Samsung's preliminary filings of 2010 to 2016 state "millions of won" while the figures are in trillions of won (e.g. revenue "57.00" for 2013Q2); corrected by a plausibility rule in 29 quarters and flagged.
- **Layout change (filings from 2025Q3).** Preliminary filings add a sign-change column after each percentage column; the prior-year figures of 2025Q3 to 2026Q2 were first misread, detected from the sample size and corrected before any interpretation.
- **Overlaps.** SK Hynix releases its results around the 25th, often one or two trading days after the first-20-days release: 13 of its 42 earnings events of 2016-2026 overlap and are excluded from the main specification.

## A6. Definitions and construction

- **Exports known at date t** (part 1). t is the first-20-days release of the quarter's last month. Known at t: full-month year-on-year growth of months 1 and 2 (ECOS index, revised series) and of the first 20 days of month 3 (KCS, first release). Quarterly growth = sum over the three months of (level a year earlier x (1 + growth)) / sum of levels a year earlier - 1.
- **Earnings measures** (first release). Revenue: year-on-year growth. Operating profit: (OP of the quarter - OP of the same quarter a year earlier) / revenue of the same quarter a year earlier, because operating profit changes sign in 16 SK Hynix quarters and 1 Samsung quarter.
- **Lead time** (V1-1). Calendar days from t to each firm's first earnings release.
- **Trading days.** Days on which any of the three series (Samsung, SK Hynix, KOSPI) has positive volume. A stock day with zero volume on a trading day is treated as missing (Samsung: 29 days, incl. 26 January to 17 March 2022); the return of that day and of the next one is left out.
- **Reaction ratio** (part 2). For each event and stock: estimation window = the 250 trading days before the event window, excluding the windows of all events; normal return = mean over that window; abnormal return = return - normal return; CAR = sum of abnormal returns from the day before to the day after the release (main window); ordinary day = mean absolute CAR over all 3-day windows of the estimation window without any event; ratio = absolute CAR / ordinary day.
- **Inference.** Reaction levels: permutation against random ordinary windows of each event's estimation period (5,000 draws). Gaps between groups: permutation of group labels. Gaps between stocks on the same events: random sign flips. 95% intervals by bootstrap over events (5,000 draws). Fixed random seed, so a re-run gives identical figures.
- **Priced-in test** (V2-5). Predicted part = fit of operating profit change on exports known at t, estimated each quarter only on quarters whose results were already public (expanding window, 8 quarters minimum); signed reaction = CAR / ordinary day on the firm's first earnings release; slope by ordinary least squares, p-value by permutation of the predicted parts.
- **Multiple testing.** Holm correction within each part: 8 tests (part 1), 22 tests (part 2).

## Figure A1. Exports known at t and earnings, four panels

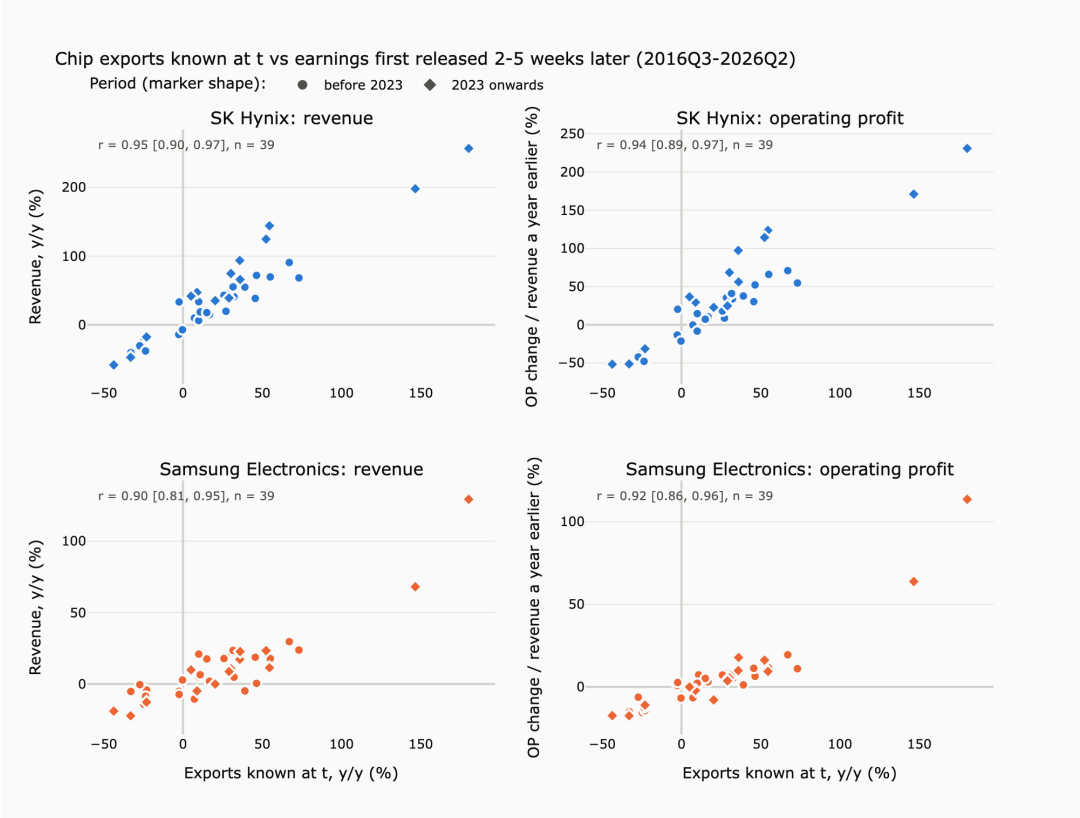



*Figure A1. Exports known at t and first-release earnings, four panels; marker shape distinguishes quarters before 2023 and from 2023.*

(moved from section 3 in the cutting pass of 2026-10-08, formerly Figure 2; renumbered 2026-10-08). The full tables of part 1 are in `appendix_tables.md` (A2).

## A4 (text). Robustness, in words

- **Part 1.** The correlations are robust to converting earnings into US dollars (changes below 0.03). Replacing ECOS by UN Comtrade data lowers them to 0.79 to 0.89, mostly because Comtrade ends in 2025 and misses the two most extreme quarters of 2026; on the same quarters, the source itself changes the correlations by about 0.02.
- **Part 2.** The conclusions on export releases hold when the reaction window is lengthened to four days (from the release day to three days after), when the release of September 2016 (whose date could not be verified) is dropped, and when overlapping releases are kept. Earnings results are less robust: with the four-day window, almost all SK Hynix earnings releases overlap with the first-20-days figure and cannot be measured, and Samsung's earnings reaction is no longer significant after Holm.
- **Priced-in test (section 5), diagnostics.** For SK Hynix, the second quarter of 2026 has a predicted part far above all others, and its results release of 29 July 2026 fell in the turbulent days before the record session of 31 July, with an abnormal return of about -33% over three days; without that quarter the slope is -0.07. For Samsung the sign of the slope depends on the first or the second quarter of 2026. The confidence intervals cannot exclude an effect of up to 1.6 ordinary days (SK Hynix) and 1.1 ordinary days (Samsung) for a typical difference (one standard deviation) in the predicted part. The conclusion does not change with revenue instead of operating profit, with UN Comtrade exports, or with earnings in US dollars.

## A7. Other limitations

- *Excluded events.* Releases whose windows overlap are left out, which removes a third of SK Hynix's results releases.
- *Daily data.* Without the time of day of each release, reactions are measured over three days.
- *Unadjusted 20-day figures.* Holidays such as the Lunar New Year shift working days between years and distort some first-20-days growth rates.
- *Prices.* Yahoo Finance data, adjusted for dividends and splits, cannot be redistributed; a re-download may give slightly different adjusted prices.

## A2. Part 1, all specifications

### V1-1. Lead time (descriptive)

| Firm | Definition | Quarters | Median (days) | Mean | Min | Max |
|---|---|---|---|---|---|---|
| SK Hynix | MAIN: first-20-days (T3) -> first earnings release | 40 | 35.0 | 35.00 | 31.0 | 44.0 |
| SK Hynix | monthly release (T3) -> first earnings release | 66 | 24.0 | 24.80 | 20.0 | 34.0 |
| SK Hynix | first-20-days (T3) -> full results | 40 | 35.0 | 35.00 | 31.0 | 44.0 |
| SK Hynix | monthly release (T3) -> full results | 66 | 24.0 | 24.80 | 20.0 | 34.0 |
| Samsung Electronics | MAIN: first-20-days (T3) -> first earnings release | 40 | 16.0 | 16.40 | 14.0 | 22.0 |
| Samsung Electronics | monthly release (T3) -> first earnings release | 67 | 6.0 | 6.10 | 3.0 | 13.0 |
| Samsung Electronics | first-20-days (T3) -> full results | 40 | 38.0 | 38.30 | 35.0 | 41.0 |
| Samsung Electronics | monthly release (T3) -> full results | 67 | 28.0 | 27.40 | 6.0 | 30.0 |

### V1-2. Correlations, 2016Q3-2026Q2

| Firm | Measure | n | r | 95% CI (Fisher) | 95% CI (block bootstrap) | p (Holm) | Expected | Verdict (Fisher / bootstrap) |
|---|---|---|---|---|---|---|---|---|
| SK Hynix | revenue y/y | 39 | 0.95 | 0.90 to 0.97 | 0.82 to 0.98 | < 0.001 | very strong | right, conclusive / right, conclusive |
| SK Hynix | OP change / revenue(t-4) | 39 | 0.94 | 0.89 to 0.97 | 0.79 to 0.98 | < 0.001 | medium or strong | wrong, conclusive / not conclusive (point estimate: wrong) |
| Samsung Electronics | revenue y/y | 39 | 0.90 | 0.81 to 0.95 | 0.65 to 0.94 | < 0.001 | medium or strong | wrong, conclusive / not conclusive (point estimate: wrong) |
| Samsung Electronics | OP change / revenue(t-4) | 39 | 0.92 | 0.86 to 0.96 | 0.82 to 0.96 | < 0.001 | weak or medium | wrong, conclusive / wrong, conclusive |

### V1-3. Comparisons of correlations

| Comparison | n | Difference in r | 95% CI (bootstrap) | p (raw) | p (Holm) | Verdict |
|---|---|---|---|---|---|---|
| SK Hynix > Samsung, revenue | 39 | 0.049 | -0.003 to 0.219 | 0.059 | 0.178 | not conclusive |
| SK Hynix > Samsung, OP | 39 | 0.014 | -0.081 to 0.056 | 0.816 | 0.816 | not conclusive |
| revenue > OP, SK Hynix | 39 | 0.009 | -0.008 to 0.035 | 0.312 | 0.624 | not conclusive |
| revenue > OP, Samsung | 39 | -0.026 | -0.203 to -0.005 | 0.025 | 0.102 | wrong |

### E-3. Before vs during the AI boom (descriptive)

| Firm | Measure | Full sample r (n) | Before 2023 r (n) | Before 2023, 95% CI |
|---|---|---|---|---|
| SK Hynix | revenue y/y | 0.95 (39) | 0.94 (25) | 0.87 to 0.97 |
| SK Hynix | OP change / revenue(t-4) | 0.94 (39) | 0.95 (25) | 0.88 to 0.98 |
| Samsung Electronics | revenue y/y | 0.90 (39) | 0.72 (25) | 0.45 to 0.87 |
| Samsung Electronics | OP change / revenue(t-4) | 0.92 (39) | 0.89 (25) | 0.77 to 0.95 |

## A3. Part 2, all specifications (main sample 2016-2026)

*V2-1: reaction ratio (1 = ordinary day). V2-2 to V2-4: gap in ordinary days. V2-5: slope of the signed reaction (ordinary days) on the predicted part.*

| Specification | n | Estimate | 95% CI | p (raw) | p (Holm, 22 tests) | Verdict |
|---|---|---|---|---|---|---|
| V2-1 first20, SK Hynix | 110 | 1.19 | 1.02 to 1.36 | 0.011 | 0.170 | descriptive |
| V2-1 first20, Samsung | 110 | 1.15 | 0.96 to 1.35 | 0.063 | 0.756 | descriptive |
| V2-1 first20, KOSPI | 113 | 1.03 | 0.87 to 1.21 | 0.750 | 1.000 | descriptive |
| V2-1 monthly, SK Hynix | 122 | 1.22 | 1.03 to 1.42 | 0.002 | 0.037 | descriptive |
| V2-1 monthly, Samsung | 121 | 1.35 | 1.13 to 1.59 | < 0.001 | 0.004 | descriptive |
| V2-1 monthly, KOSPI | 122 | 1.25 | 1.03 to 1.52 | < 0.001 | 0.018 | descriptive |
| V2-1 earnings_samsung, Samsung | 43 | 1.50 | 1.19 to 1.86 | < 0.001 | 0.012 | right, conclusive |
| V2-1 earnings_samsung, KOSPI | 43 | 1.28 | 0.96 to 1.61 | 0.038 | 0.489 | right, conclusive |
| V2-1 earnings_sk, SK Hynix | 29 | 1.63 | 1.23 to 2.09 | < 0.001 | 0.012 | not conclusive (point estimate: right) |
| V2-1 earnings_sk, KOSPI | 29 | 1.27 | 0.82 to 1.79 | 0.073 | 0.801 | right, conclusive |
| V2-2 first20 > monthly, SK Hynix | 110 vs 122 | -0.03 | -0.30 to 0.23 | 0.827 | 1.000 | wrong |
| V2-2 first20 > monthly, Samsung | 110 vs 121 | -0.20 | -0.51 to 0.10 | 0.182 | 1.000 | wrong |
| V2-2 first20 > monthly, KOSPI | 113 vs 122 | -0.22 | -0.53 to 0.07 | 0.166 | 1.000 | wrong |
| V2-3 monthly before 2016 > after, SK Hynix | 139 vs 122 | -0.25 | -0.48 to -0.04 | 0.029 | 0.403 | wrong (not interpretable: rule-based dates) |
| V2-3 monthly before 2016 > after, Samsung | 149 vs 121 | -0.45 | -0.70 to -0.19 | < 0.001 | 0.008 | wrong (not interpretable: rule-based dates) |
| V2-3 monthly before 2016 > after, KOSPI | 147 vs 122 | -0.15 | -0.44 to 0.11 | 0.266 | 1.000 | wrong (not interpretable: rule-based dates) |
| V2-4 monthly, SK Hynix > Samsung | 121 | -0.14 | -0.36 to 0.07 | 0.185 | 1.000 | wrong |
| V2-4 monthly, Samsung > KOSPI | 120 | 0.08 | -0.11 to 0.28 | 0.418 | 1.000 | wrong |
| V2-4 first20, SK Hynix > Samsung | 110 | 0.04 | -0.18 to 0.25 | 0.719 | 1.000 | wrong |
| V2-4 first20, Samsung > KOSPI | 110 | 0.11 | -0.07 to 0.29 | 0.248 | 1.000 | wrong |
| V2-5 priced-in test, SK Hynix | 22 | -1.90 | -2.71 to 1.74 | 0.025 | 0.381 | consistent with priced in, not a proof |
| V2-5 priced-in test, Samsung | 31 | -1.96 | -7.09 to 6.18 | 0.434 | 1.000 | consistent with priced in, not a proof |

# A4. Robustness

## Part 1 (estimates)

| Specification | Main | Comtrade for months 1-2 | Earnings in USD |
|---|---|---|---|
| V1-2 SK Hynix, OP change / revenue(t-4) | 0.939 | 0.863 | 0.941 |
| V1-2 SK Hynix, revenue y/y | 0.948 | 0.875 | 0.957 |
| V1-2 Samsung Electronics, OP change / revenue(t-4) | 0.924 | 0.893 | 0.936 |
| V1-2 Samsung Electronics, revenue y/y | 0.899 | 0.789 | 0.926 |
| V1-3 SK Hynix > Samsung, OP | 0.014 | -0.030 | 0.006 |
| V1-3 SK Hynix > Samsung, revenue | 0.049 | 0.086 | 0.031 |
| V1-3 revenue > OP, SK Hynix | 0.009 | 0.012 | 0.016 |
| V1-3 revenue > OP, Samsung | -0.026 | -0.103 | -0.010 |

## Part 2 (estimates; * = significant at 5% after Holm within the variant)

| Specification | main | window [0,+3] | market model, SOX | without 2026-07-31 | without unverified 2016-09 | overlaps included | V2-5 on revenue | V2-5 exports from Comtrade | V2-5 earnings in USD |
|---|---|---|---|---|---|---|---|---|---|
| V2-1 first20, SK Hynix | 1.19 | 1.25 | 1.30* | 1.19 | 1.20 | 1.18 | 1.19 | 1.19 | 1.19 |
| V2-1 first20, Samsung | 1.15 | 1.02 | 1.20 | 1.15 | 1.16 | 1.12 | 1.15 | 1.15 | 1.15 |
| V2-1 first20, KOSPI | 1.03 | 1.09 | 1.13 | 1.03 | 1.03 | 1.01 | 1.03 | 1.03 | 1.03 |
| V2-1 monthly, SK Hynix | 1.22* | 1.29* | 1.23* | 1.20 | 1.22* | 1.22* | 1.22* | 1.22* | 1.22* |
| V2-1 monthly, Samsung | 1.35* | 1.46* | 1.39* | 1.33* | 1.35* | 1.33* | 1.35* | 1.35* | 1.35* |
| V2-1 monthly, KOSPI | 1.25* | 1.51* | 1.33* | 1.22 | 1.25* | 1.24* | 1.25* | 1.25* | 1.25* |
| V2-1 earnings_samsung, Samsung | 1.50* | 1.38 | 1.48* | 1.50* | 1.50* | 1.50* | 1.50* | 1.50* | 1.50* |
| V2-1 earnings_samsung, KOSPI | 1.28 | 1.28 | 1.22 | 1.28 | 1.28 | 1.28 | 1.28 | 1.28 | 1.28 |
| V2-1 earnings_sk, SK Hynix | 1.63* |  | 1.59* | 1.63* | 1.63* | 1.48* | 1.63* | 1.63* | 1.63* |
| V2-1 earnings_sk, KOSPI | 1.27 |  | 1.25 | 1.27 | 1.27 | 1.18 | 1.27 | 1.27 | 1.27 |
| V2-2 first20 > monthly, SK Hynix | -0.03 | -0.04 | 0.08 | -0.01 | -0.02 | -0.04 | -0.03 | -0.03 | -0.03 |
| V2-2 first20 > monthly, Samsung | -0.20 | -0.44 | -0.19 | -0.18 | -0.20 | -0.21 | -0.20 | -0.20 | -0.20 |
| V2-2 first20 > monthly, KOSPI | -0.22 | -0.41 | -0.20 | -0.20 | -0.22 | -0.23 | -0.22 | -0.22 | -0.22 |
| V2-3 monthly before 2016 > after, SK Hynix | -0.25 | -0.32 | -0.22 | -0.24 | -0.25 | -0.27 | -0.25 | -0.25 | -0.25 |
| V2-3 monthly before 2016 > after, Samsung | -0.45* | -0.30 | -0.47* | -0.42* | -0.45* | -0.42* | -0.45* | -0.45* | -0.45* |
| V2-3 monthly before 2016 > after, KOSPI | -0.15 | -0.42 | -0.25 | -0.13 | -0.15 | -0.17 | -0.15 | -0.15 | -0.15 |
| V2-4 monthly, SK Hynix > Samsung | -0.14 | -0.16 | -0.17 | -0.14 | -0.14 | -0.12 | -0.14 | -0.14 | -0.14 |
| V2-4 monthly, Samsung > KOSPI | 0.08 | -0.04 | 0.04 | 0.09 | 0.08 | 0.07 | 0.08 | 0.08 | 0.08 |
| V2-4 first20, SK Hynix > Samsung | 0.04 | 0.22 | 0.11 | 0.04 | 0.04 | 0.06 | 0.04 | 0.04 | 0.04 |
| V2-4 first20, Samsung > KOSPI | 0.11 | -0.03 | 0.05 | 0.11 | 0.12 | 0.09 | 0.11 | 0.11 | 0.11 |
| V2-5 priced-in test, SK Hynix | -1.90 |  | -1.75 | -1.90 | -1.90 | -1.30 | -1.80 | -0.17 | -1.88 |
| V2-5 priced-in test, Samsung | -1.96 | -3.90 | -1.44 | -1.96 | -1.96 | -1.96 | -1.51 | -1.95 | -1.87 |

*Empty cell: not computable (fewer than 10 events, e.g. SK Hynix earnings with the [0,+3] window, almost all overlapping the first-20-days release). V2-5 with Comtrade exports: the SK Hynix slope falls to about -0.2 because Comtrade ends in 2025 and drops 2026Q2, the quarter that drives the main estimate (see section 5).*

# A5. Exploratory specifications (not in the Holm families)

| Specification | n | Estimate | 95% CI | p (raw) | Verdict | Note |
|---|---|---|---|---|---|---|
| E-1 earnings reaction vs lead time, SK Hynix | 27 | 0.13 | -0.17 to 0.48 | 0.354 | not conclusive | lead 31-37 days, sd 1.7 |
| E-1 earnings reaction vs lead time, Samsung | 40 | 0.05 | -0.15 to 0.36 | 0.612 | not conclusive | lead 14-22 days, sd 1.9 |
| E-2 monthly, top 20% surprises vs others, SK Hynix | 23 vs 89 | -0.01 | -0.77 to 0.65 | 0.959 | not conclusive | top 20%: mean signed -0.07 sigma; others -0.06; absolute surprise cut 11.9 pp |
| E-2 monthly, top 20% surprises vs others, Samsung | 23 vs 88 | -0.45 | -1.29 to 0.30 | 0.189 | wrong | top 20%: mean signed -0.48 sigma; others -0.03; absolute surprise cut 11.8 pp |
| E-2 monthly, top 20% surprises vs others, KOSPI | 23 vs 89 | -0.54 | -1.64 to 0.35 | 0.121 | wrong | top 20%: mean signed -0.55 sigma; others -0.00; absolute surprise cut 11.8 pp |
| E-2 first-20-days, SK Hynix | 0 | | | | not computable (no free expectation) | |
| E-2 first-20-days, Samsung | 0 | | | | not computable (no free expectation) | |
| E-2 first-20-days, KOSPI | 0 | | | | not computable (no free expectation) | |